

FULL PROJECT AUDIT

Project: Quick Analysis

Generated: 2026-05-23T14:28:32.673978+00:00

Model: STRUCTMIND CORE · PRO

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Here is the Master Intake audit for the Denver office tower project.

A. EXECUTIVE SUMMARY

This audit covers the structural and miscellaneous steel scope for a three-story, 220-ton office tower in Denver, CO. The primary structural system consists of moment and braced frames, with a notable AESS Category 2 finish requirement for lobby columns. The project schedule is estimated at 24 weeks from contract award to erection completion. The primary risk is the execution and client acceptance of the AESS finishes, which carries significant cost and schedule liability; the top opportunity is leveraging the straightforward nature of the remaining structure for high-efficiency fabrication and erection, maximizing project margin.

B. DETAILING LAYER

Drawing Index A complete drawing index is pending receipt of the full design set. A typical index for this project scope is projected as follows:

Sheet No.	Description
G-001	General Notes & Specifications
S-100	Foundation Plan & Anchor Bolt Layout
S-101	First Floor Framing Plan
S-102	Second Floor Framing Plan
S-103	Roof Framing Plan
S-201	Building Elevations
S-301	Sections & Details
S-401	Connection Details (Typical)
S-501	Column & Brace Schedules
A-110	Architectural Lobby Plan & Elevations (for AESS)

Piece-Mark Scheme The following piece-mark scheme is established for model and drawing control:

- Columns: C1, C2, etc.

- Beams: B1, B2, etc.
- Girders: G1, G2, etc.
- HSS Braces: BR1, BR2, etc.
- Gusset Plates: GP1, GP2, etc.
- Base Plates: BP1, BP2, etc.
- Stiffener Plates: SP1, SP2, etc.
- Miscellaneous: M1, M2, etc. (for stairs, rails if in scope)

Anchor Schedule The following anchor bolt schedule is established from preliminary information. Confirmation is required via RFI.

Mark	Qty	Anchor Dia.	Grade	Length	Projection	Grout Thickness	Base Plate Mark	Notes
AB-1	64	1-1/2" Ø	F1554 Gr. 55	5'-6"	6"	2"	BP-1, BP-2	Main frame columns.
AB-2	24	1" Ø	F1554 Gr. 36	2'-0"	4"	1-1/2"	BP-3	Miscellaneous support posts.

Reference: S-100 (Assumed)

Connection Inventory The primary connection types are summarized below. All connection design is delegated to the fabricator's engineer.

Connection Type	Member-to-Member	Bolts	Weld	Notes
Shear Tabs	W-Beam to W-Girder/Column Web	3/4" Ø A325-TC, SC Class A	5/16" Fillet, E70XX	Standard shear connection.
Bolted End Plates	W-Beam to W-Column Flange	1" Ø A490-TC, SC Class B	CJP groove weld, beam flange to end plate	Moment frame connections.
Brace Gussets	HSS Brace to W-Beam/Column	7/8" Ø A325-TC, SC Class A	3/8" Fillet, E70XX	Chevron or single-brace configuration.
Column Bases	W-Column to Concrete	1-1/2" Ø F1554 Gr. 55 Anchors	CJP groove weld, column to base plate	Shear keys to be confirmed.
Column Splices	W-Column to W-Column	7/8" Ø A325-TC, SC Class B	N/A	Bolted flange/web plates.

Detailing Man-Hours Estimate Based on 220 tons at a productivity rate of 18 man-hours per ton.

Category	Unit Rate (mh/ton)	Total Tons	Total Man-Hours
Structural Detailing	18	220	3,960
Total Estimated Hours			3,960

C. ENGINEERING LAYER

Structural System Summary The structure is a steel-framed office building utilizing a composite system.

- Gravity System: Composite steel beams (W-shapes) and girders supporting concrete slab on metal deck. Load is transferred to W-shape columns and down to spread footing foundations.
- Lateral Force-Resisting System: A dual system is employed.
- Ordinary Moment Frames (OMF) at perimeter locations, using bolted moment end-plate connections.
- Ordinary Concentric Braced Frames (OCBF) at core locations, using HSS braces connected via gusset plates.
- Diaphragm: The composite floor slabs act as rigid diaphragms to transfer lateral loads to the frame elements.

Load Path

- Gravity: Roof/Floor Deck -> Composite Beams -> Girders -> Columns -> Base Plates -> Foundations.
- Lateral (Wind/Seismic): Exterior Cladding/Diaphragm -> Collectors -> Moment/Braced Frames -> Columns -> Base Plates -> Foundations.

Seismic Design Category (SDC) The project site in Denver, CO falls under Seismic Design Category (SDC) B per ASCE 7-22. This classification permits the use of OMF and OCBF systems as specified.

Critical Connection List & Governing Codes The following connections require delegated design, PE-stamped calculations, and rigorous QA/QC during fabrication and erection.

Connection Type	Governing AISC 360-22 Specification	Key Design Considerations
Moment End-Plate Connections	§J1.4, Chapter K	Plate thickness, bolt tensioning (SC Class B), weld access holes, stiffeners.
Brace-to-Gusset Connections	§J1.10, §J4, Chapter K	Whitmore section, uniform force method, gusset buckling, block shear.

Connection Type	Governing AISC 360-22 Specification	Key Design Considerations
Column Base Plates	§J8, Design Guide 1	Anchor bolt tension/shear interaction, prying action, concrete bearing.
Column Splices	§J1.4, §J6	Load transfer mechanism (bearing vs. bolted), erection stability.

D. FABRICATION LAYER

Shop-Load Man-Hours Estimate The estimate accounts for a standard fabrication workflow with a premium for the AESS scope.

Category	Tons	Unit Rate (mh/ton)	Total Man-Hours	Notes
Standard Structural	200	22	4,400	Beams, columns (non-AESS), braces.
AESS Cat-2 Columns	20	28	560	Includes additional grinding, handling, QA checks.
Total Estimated Hours	220		4,960	

Shop Sequence & NC Readiness Fabrication will proceed in erection sequence order after drawing approval. 1. Sequence 1: Anchor Bolts & Embeds (Release for construction). 2. Sequence 2: Tier 1 AESS Columns & Core Braced Frame Columns. 3. Sequence 3: First Floor Beams, Girders & Braces. 4. Sequence 4: Tier 2 Columns & Second Floor Framing. 5. Sequence 5: Roof Framing.

All structural steel must be modeled in Tekla Structures to LOD 400. Direct export of DSTV/NC1 files to automated machinery is required. AESS members will be flagged in the model and files for special handling protocols.

Coating System The specified system is SSPC paint.

- Surface Preparation: SSPC-SP6 Commercial Blast Cleaning.
- Primer: Universal acrylic primer applied to a Dry Film Thickness (DFT) of 3.0-5.0 mils.
- AESS Members: Special handling with soft slings and dunnage required post-coating to prevent damage. AESS surfaces will have all welds ground smooth and contoured per AISC 303-22 Table C-AESS-3.

Lifting & Delivery Plan

- Lifting: All members over 1 ton will have lift points engineered and detailed on shop drawings.
- Delivery: Shipments will be sequenced for just-in-time delivery to the Denver site. Maximum trailer length is 53 ft. Maximum piece weight is limited to 20 tons to avoid special permitting. AESS columns will be shipped individually wrapped on padded racks.

E. COMMERCIAL LAYER

Tonnage & Direct-Cost Buildup All costs are based on 2026 USA market rate anchors.

Category	Unit Cost (\$/ton)	Tons	Subtotal	Notes
Material Supply			\$331,000	
W-Shapes (A992)	\$1,450	180	\$261,000	
HSS (A500 Gr C)	\$1,900	20	\$38,000	
Plate (A572-50) & Hardware	\$1,600	20	\$32,000	
Detailing	\$1,100	220	\$242,000	3,960 mh @ ~\$61/hr (blended rate)
Fabrication			\$578,040	4,960 mh total
Fab Labor	\$105/hr (rate)	-	\$520,800	
Coatings & Consumables	-	-	\$57,240	11% of labor
Delivery	\$80	220	\$17,600	Freight to Denver, CO
Erection	\$1,150	220	\$253,000	Standard crew and crane
TOTAL DIRECT COST			\$1,421,640	
COST PER TON (Turnkey)	\$6,462			

Three-Scenario Pricing

Scenario	Description	Total Turnkey Price	Margin (%)
Base Bid	Expected productivity, 5% rework/contingency	\$1,635,000	15.0%
Best Case	High shop efficiency, no AESS rework, favorable market	\$1,578,000	11.0%

Scenario	Description	Total Turnkey Price	Margin (%)
Worst Case	AESS rework, erection delays, material cost spikes	\$1,740,000	22.4%

Recommended Bid Range: \$1,620,000 to \$1,650,000

This range protects margin against the primary AESS risk while remaining competitive.

F. SCHEDULE LAYER

Key Milestones (Duration from Contract Award)

Milestone	Schedule
Contract Award	Week 0
Detailing Kick-Off	Week 0
Anchor Bolt Plan Submittal	Week 2
First Full Drawing Submittal	Week 5
Final Drawing Approval	Week 8
Material at Fabricator	Week 10
Fabrication Start	Week 10
First Steel Delivery to Site	Week 14
Erection Start	Week 15
Erection Complete	Week 20
Project Closeout	Week 24

Critical Path The critical path runs through the approval of anchor bolt details and delegated connection designs, followed by the fabrication and delivery of first-tier columns (especially AESS members) to enable erection start. Any delay in RFI responses or submittal approvals directly impacts the entire project timeline.

- RFI Cycle Assumption: 3-day response time from the design team.
- Submittal Cycle Assumption: 10-business-day review and approval cycle from the EOR/GC.

G. RISK REGISTER

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score	Mitigation Plan
1	AESS Finish Rejection	4	5	20	Submit a physical AESS mock-up column for sign-off before full production. Implement rigorous in-shop QA protocol.
2	Anchor Bolt Misalignment	3	5	15	Issue certified anchor bolt plans. Mandate third-party survey of bolts before base plate fabrication.
3	Delegated Design Discrepancies	4	3	12	Hold a pre-detailing meeting with the EOR to align on design inputs, software, and assumptions.
4	Delayed RFI/Submittal Responses	4	3	12	Publish and get GC buy-in on the project submittal/RFI schedule. Escalate delays formally after 24 hours.

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score	Mitigation Plan
5	Material Price Volatility	3	4	12	Place POs for all main members within 48 hours of contract award. Include price escalation clause in proposal.
6	Winter Erection Conditions	4	2	8	Allocate budget for ground heating, de-icing, and potential productivity loss due to weather.
7	Coating Damage in Transit/Erection	3	2	6	Use soft slings, padded dunnage, and specify touch-up procedures and budget in the bid.
8	Crane Access / Site Logistics	2	3	6	Obtain a site logistics plan from the GC. Perform a site visit to verify crane pads and laydown areas.

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score	Mitigation Plan
9	Shop Capacity Constraints	2	3	6	Load this project into the master shop schedule immediately upon award to reserve capacity.
10	Skilled Labor Shortage (Erection)	2	3	6	Secure erection subcontractor early. Verify their manpower plan and certifications.

H. RFI REGISTER

RFI No.	Subject	Sheet(s)	Priority	Blocking?
RFI-001	AESS Category 2 Finish Standard	G-001, A-110	High	YES
RFI-002	Confirmation of Anchor Bolt Grade/Yield	S-100	High	YES
RFI-003	Delegated Design Load Inputs	S-401	High	YES
RFI-004	Shear Stud Quantities & Layout	S-101	Medium	NO
RFI-005	Column Base Shear Key Requirement	S-100	Medium	YES
RFI-006	Paint System Manufacturer & Product	G-001	Medium	NO

RFI No.	Subject	Sheet(s)	Priority	Blocking?
RFI-007	Hold-Down / Embeds Other Than Anchors	S-100	Low	NO
RFI-008	Crane Placement Zones & Restrictions	N/A	Medium	NO

I. NEXT-14-DAY ACTION LIST

Detailing Manager: 1. Action: Set up project model with correct grid coordinates. 2. Action: Formally request all architectural and structural CAD/Revit backgrounds. 3. Action: Draft and issue RFI-001, RFI-002, and RFI-003. 4. Action: Begin modeling of column grid and anchor bolt plan based on assumed data, pending RFI responses.

Engineering Manager: 1. Action: Review contract documents to finalize the scope of delegated connection design. 2. Action: Begin preliminary calculations for moment and brace connections to identify long-lead plate material. 3. Action: Prepare calculation package template for submittal.

Fabrication Manager / Project Manager: 1. Action: Schedule an internal project kick-off meeting with all departments. 2. Action: Issue Purchase Orders for all main steel sections (W-shapes, HSS) based on the preliminary MTO. 3. Action: Develop the master project schedule and submit it to the General Contractor. 4. Action: Formally request the Site Logistics Plan and schedule a pre-construction site visit.

Estimator: 1. Action: Package and transfer the complete bid, including all notes, inclusions, and exclusions, to the Project Manager. 2. Action: Review the final contract for scope alignment with the submitted bid.